



Lecture 2: Mathematical Models

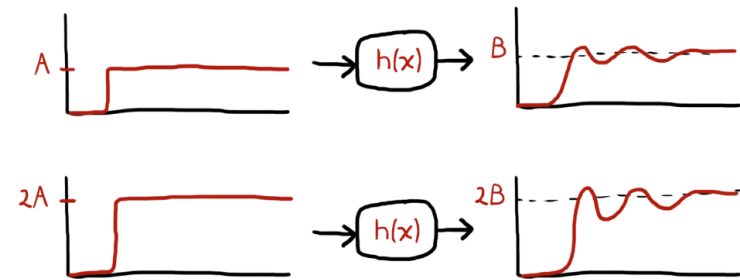
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SIST 1D#206



Linear System

- Linearity is defined in terms of the input $u(t)$ and output $y(t)$
 - Also known as excitation and response.
- If a system satisfies **homogeneity** and **superposition**, it is a linear system.
- If a linear system further sanctifies the property of **time-invariance**, it is then called an LTI system.
 - Linearity + Time-invariance = **sinusoidal fidelity**
 - Linear system is a wider concept than *linear time-invariant (LTI)* system, also including *linear time-varying* system

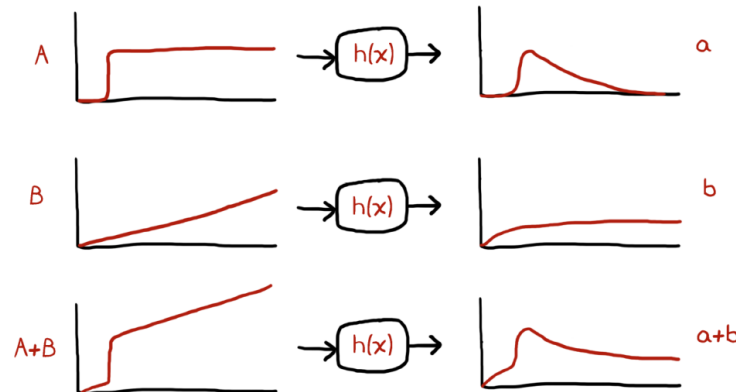
Homogeneity



Homogeneity

$$h(ax) = a \cdot h(x)$$

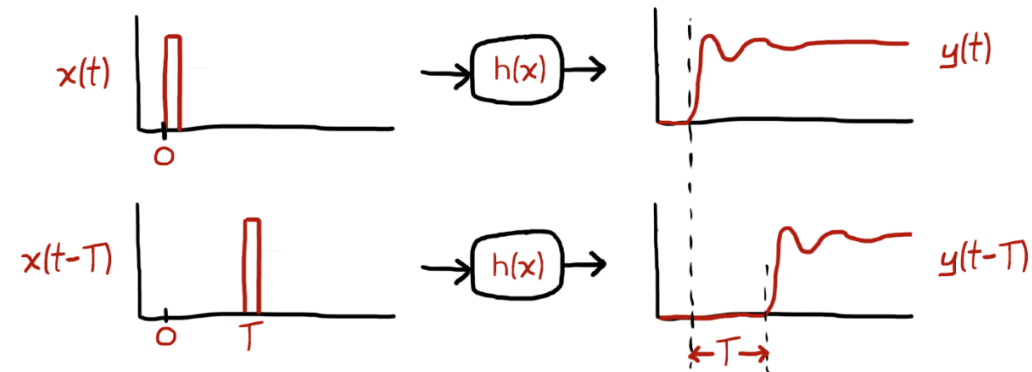
Superposition



Superposition

$$h(x_1) + h(x_2) = h(x_1 + x_2)$$

Time Invariance



- Linear time-invariant operator

LTI Allowable operations:

Multiply or divide the input by a constant

$$a \cdot x(t) \quad \frac{1}{a} \cdot x(t)$$

Integrate or differentiate the input

$$\int x(t) dt \quad \frac{dx(t)}{dt}$$

Add or subtract multiple inputs

$$x_1(t) + x_2(t) \quad x_1(t) - x_2(t)$$

- Generally speaking, a system is also an operator (we are not ready to see this yet)

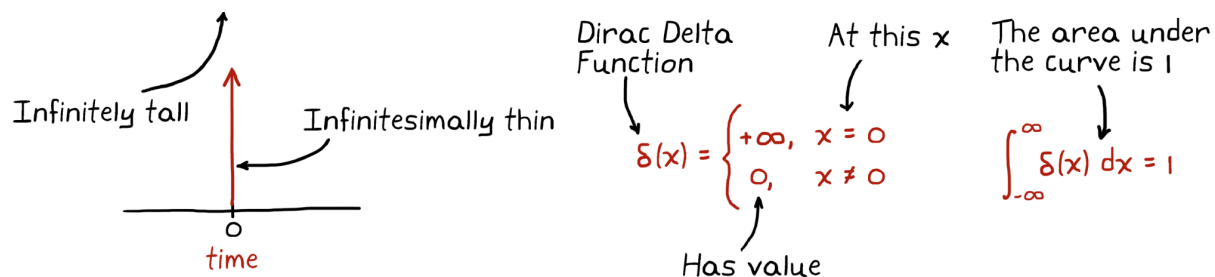
True or False:

The differential equation $\frac{d}{dt}y = -ty + u$ describes a linear system.



Laplace Transform

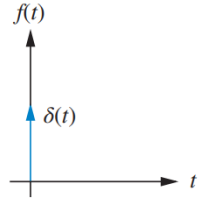
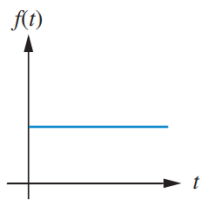
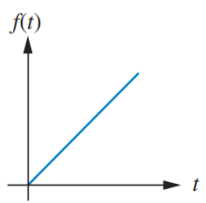
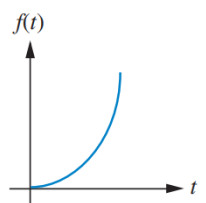
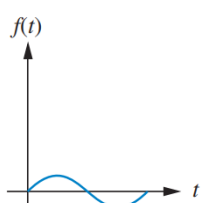
Dirac delta function $\delta(t)$



Heaviside step function $u(t)$

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t < 0 \end{cases}$$

TABLE 1.1 Test waveforms used in control systems

Input	Function	Description	Sketch	Use
Impulse	$\delta(t)$	$\delta(t) = \infty$ for $0- < t < 0+$ $= 0$ elsewhere $\int_{0-}^{0+} \delta(t) dt = 1$		Transient response Modeling
Step	$u(t)$	$u(t) = 1$ for $t > 0$ $= 0$ for $t < 0$		Transient response Steady-state error
Ramp	$tu(t)$	$tu(t) = t$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Parabola	$\frac{1}{2}t^2u(t)$	$\frac{1}{2}t^2u(t) = \frac{1}{2}t^2$ for $t \geq 0$ $= 0$ elsewhere		Steady-state error
Sinusoid	$\sin \omega t$			Transient response Modeling Steady-state error

- Laplace transform is an integral transform with a fast decaying kernel function,
 - because the most important thing for an integral is to exist

$$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$$

Kernel is: $K(s, t) = e^{-st}$

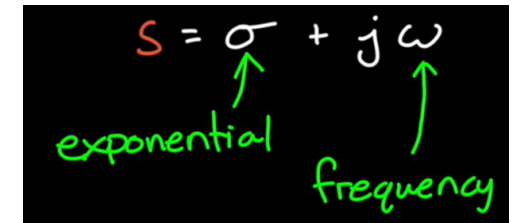


TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t) \quad \begin{matrix} u(t) = 1 & t > 0 \\ = 0 & t < 0 \end{matrix}$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

Example 2.1

Laplace Transform of a Time Function

PROBLEM: Find the Laplace transform of $f(t) = Ae^{-at}u(t)$.

SOLUTION: Since the time function does not contain an impulse function, we can replace the lower limit of Eq. (2.1) with 0. Hence,

$$\begin{aligned} F(s) &= \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} Ae^{-at}e^{-st} dt = A \int_0^{\infty} e^{-(s+a)t} dt \\ &= -\frac{A}{s+a} e^{-(s+a)t} \Big|_{t=0}^{\infty} = \frac{A}{s+a} \end{aligned} \quad (2.3)$$

❑ Inverse Laplace transform is often done with further help of the following table

Example 2.2

Inverse Laplace Transform

PROBLEM: Find the inverse Laplace transform of $F_1(s) = 1/(s + 3)^2$.

SOLUTION: For this example we make use of the frequency shift theorem, Item 4 of Table 2.2, and the Laplace transform of $f(t) = tu(t)$, Item 3 of Table 2.1. If the inverse transform of $F(s) = 1/s^2$ is $tu(t)$, the inverse transform of $F(s + a) = 1/(s + a)^2$ is $e^{-at}tu(t)$. Hence, $f_1(t) = e^{-3t}tu(t)$.

TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t - T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{k-1}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²

□ The two tables can be used to derive for more complicated results, e.g.,

$$F_1(s) = \frac{s^3 + 2s^2 + 6s + 7}{s^2 + s + 5}$$

$$F_1(s) = s + 1 + \frac{2}{s^2 + s + 5}$$

$$f_1(t) = \frac{d\delta(t)}{dt} + \delta(t) + \mathcal{L}^{-1} \left[\frac{2}{s^2 + s + 5} \right]$$

TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
4.	$\mathcal{L}[e^{-at}f(t)] = F(s + a)$	Frequency shift theorem

$$\mathcal{L}[Ae^{-at} \cos \omega t] = \frac{A(s + a)}{(s + a)^2 + \omega^2}$$

- Laplace transform converts a differential equation into algebraic equation for its **differentiation theorem**

TABLE 2.2 Laplace transform theorems

Item no.	Theorem		Name
7.	$\mathcal{L}\left[\frac{df}{dt}\right]$	$= sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right]$	$= s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right]$	$= s^nF(s) - \sum_{k=1}^n s^{n-k}f^{k-1}(0-)$	Differentiation theorem

Example 2.3

Laplace Transform Solution of a Differential Equation

PROBLEM: Given the following differential equation, solve for $y(t)$ if all initial conditions are zero. Use the Laplace transform.

$$\frac{d^2y}{dt^2} + 12\frac{dy}{dt} + 32y = 32u(t) \quad (2.14)$$

Laplace Transform



PROBLEM: Find the Laplace transform of $f(t) = te^{-5t}$.

ANSWER:



TABLE 2.1 Laplace transform table

Item no.	$f(t)$	$F(s)$
1.	$\delta(t)$	1
2.	$u(t) \quad \begin{matrix} u(t) = 1 & t > 0 \\ = 0 & t < 0 \end{matrix}$	$\frac{1}{s}$
3.	$tu(t)$	$\frac{1}{s^2}$
4.	$t^n u(t)$	$\frac{n!}{s^{n+1}}$
5.	$e^{-at} u(t)$	$\frac{1}{s+a}$
6.	$\sin \omega t u(t)$	$\frac{\omega}{s^2 + \omega^2}$
7.	$\cos \omega t u(t)$	$\frac{s}{s^2 + \omega^2}$

PROBLEM: Find the inverse Laplace transform of $F(s) = 10/[s(s+2)(s+3)^2]$.

ANSWER:



TABLE 2.2 Laplace transform theorems

Item no.	Theorem	Name
1.	$\mathcal{L}[f(t)] = F(s) = \int_{0-}^{\infty} f(t)e^{-st} dt$	Definition
2.	$\mathcal{L}[kf(t)] = kF(s)$	Linearity theorem
3.	$\mathcal{L}[f_1(t) + f_2(t)] = F_1(s) + F_2(s)$	Linearity theorem
4.	$\mathcal{L}[e^{-at}f(t)] = F(s+a)$	Frequency shift theorem
5.	$\mathcal{L}[f(t-T)] = e^{-sT}F(s)$	Time shift theorem
6.	$\mathcal{L}[f(at)] = \frac{1}{a}F\left(\frac{s}{a}\right)$	Scaling theorem
7.	$\mathcal{L}\left[\frac{df}{dt}\right] = sF(s) - f(0-)$	Differentiation theorem
8.	$\mathcal{L}\left[\frac{d^2f}{dt^2}\right] = s^2F(s) - sf(0-) - f'(0-)$	Differentiation theorem
9.	$\mathcal{L}\left[\frac{d^nf}{dt^n}\right] = s^nF(s) - \sum_{k=1}^n s^{n-k}f^{(k-1)}(0-)$	Differentiation theorem
10.	$\mathcal{L}\left[\int_{0-}^t f(\tau)d\tau\right] = \frac{F(s)}{s}$	Integration theorem
11.	$f(\infty) = \lim_{s \rightarrow 0} sF(s)$	Final value theorem ¹
12.	$f(0+) = \lim_{s \rightarrow \infty} sF(s)$	Initial value theorem ²



Transfer Function

- General case of a high order differential equation:

$$a_n \frac{d^n c(t)}{dt^n} + a_{n-1} \frac{d^{n-1} c(t)}{dt^{n-1}} + \cdots + a_0 c(t) = b_m \frac{d^m r(t)}{dt^m} + b_{m-1} \frac{d^{m-1} r(t)}{dt^{m-1}} + \cdots + b_0 r(t) \quad (2.50)$$

- Convert it into s-domain:

$$\begin{aligned} a_n s^n C(s) + a_{n-1} s^{n-1} C(s) + \cdots + a_0 C(s) + \text{initial condition} \\ \text{terms involving } c(t) \\ = b_m s^m R(s) + b_{m-1} s^{m-1} R(s) + \cdots + b_0 R(s) + \text{initial condition} \\ \text{terms involving } r(t) \end{aligned}$$

- Assume zero initial conditions we get transfer function

$$\frac{C(s)}{R(s)} = G(s) = \frac{(b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0)}{(a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0)}$$

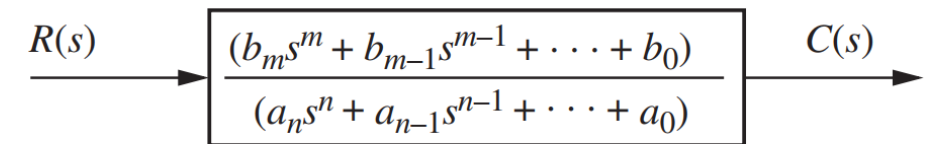


FIGURE 2.2 Block diagram of a transfer function

PROBLEM: Find the transfer function, $G(s) = C(s)/R(s)$, corresponding to the differential equation $\frac{d^3 c}{dt^3} + 3\frac{d^2 c}{dt^2} + 7\frac{dc}{dt} + 5c = \frac{d^2 r}{dt^2} + 4\frac{dr}{dt} + 3r$.

ANSWER:

PROBLEM: Find the differential equation corresponding to the transfer function,

$$G(s) = \frac{2s + 1}{s^2 + 6s + 2}$$

ANSWER:

PROBLEM: Find the ramp response for a system whose transfer function is

$$G(s) = \frac{s}{(s + 4)(s + 8)}$$

ANSWER:



System Examples

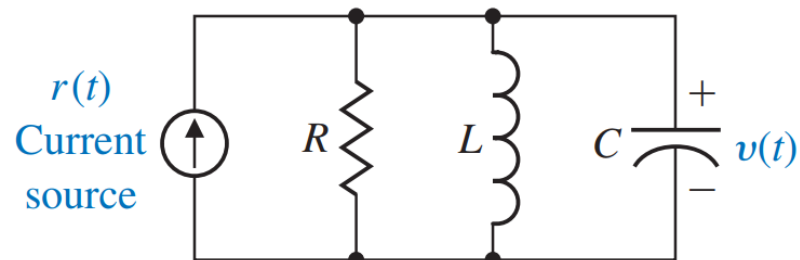
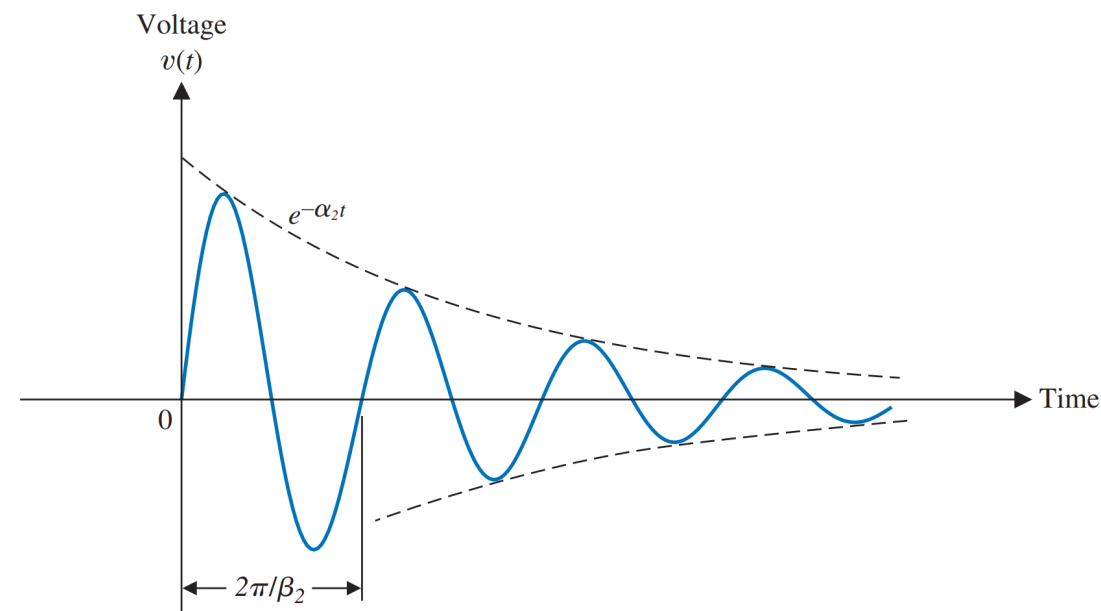


FIGURE 2.3
RLC circuit.

$$\frac{v(t)}{R} + C \frac{dv(t)}{dt} + \frac{1}{L} \int_0^t v(t) dt = r(t)$$

$$v(t) = K_2 e^{-\alpha_2 t} \cos(\beta_2 t + \theta_2)$$

FIGURE 2.4
Typical voltage response for an RLC circuit.



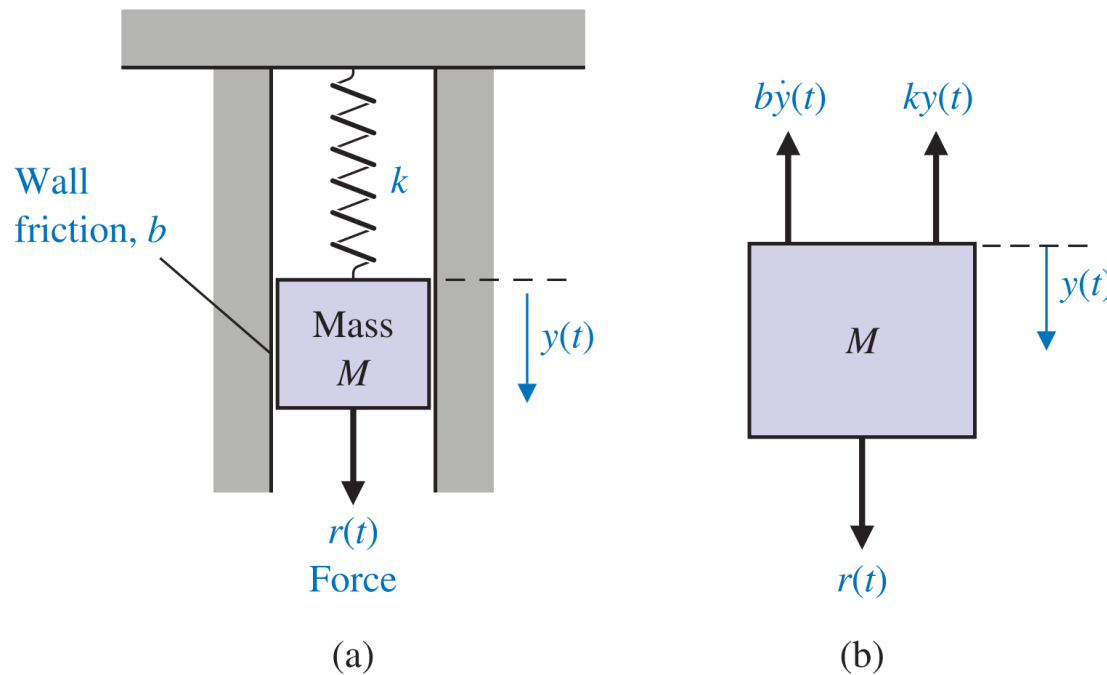


FIGURE 2.2
(a) Spring-mass-damper system.
(b) Free-body diagram.

$$M \frac{d^2 y(t)}{dt^2} + b \frac{dy(t)}{dt} + ky(t) = r(t)$$

$$y(t) = K_1 e^{-\alpha_1 t} \sin(\beta_1 t + \theta_1)$$

- ❑ Newtonian mechanics
- ❑ Faraday's law of induction
 - With a little bit of math of total differential and derivative laws
- ❑ Ampere force

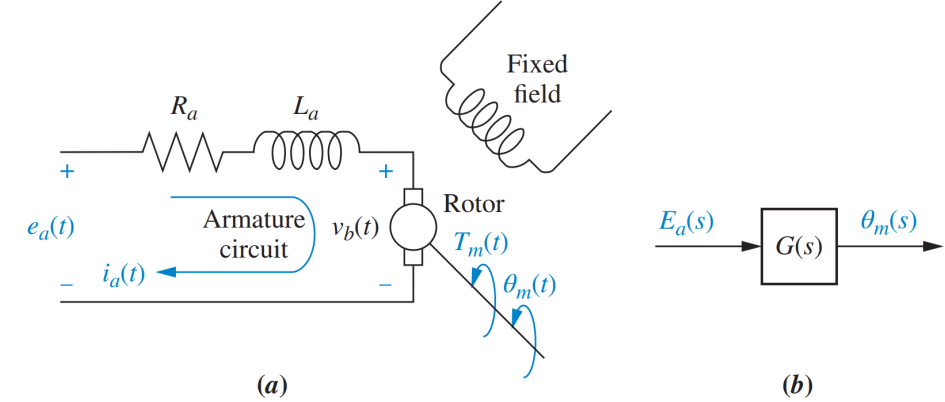


FIGURE 2.35 DC motor: **a.** schematic;¹² **b.** block diagram

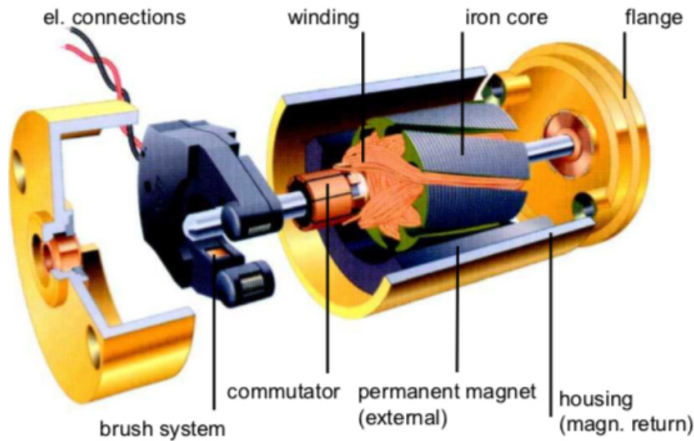
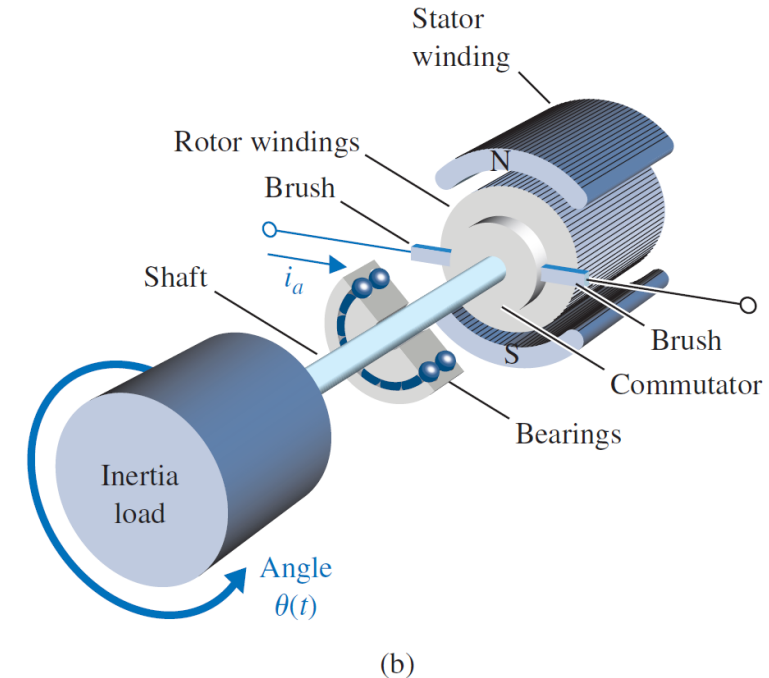
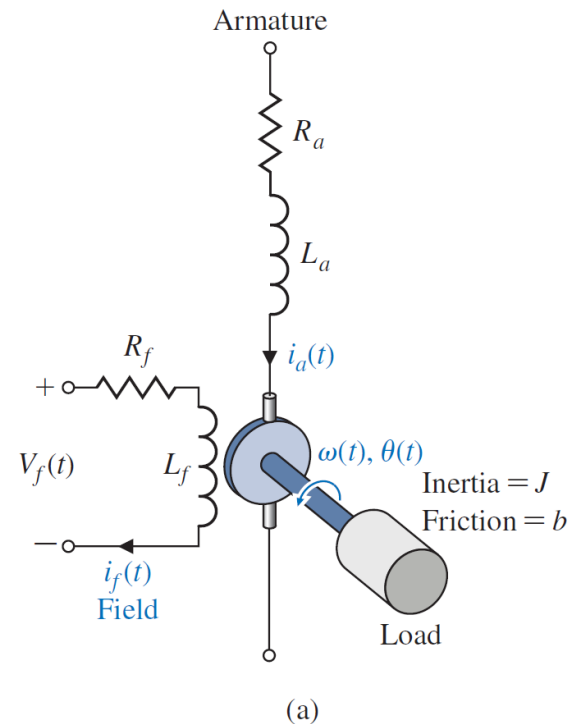


FIGURE 2.18
A DC motor
(a) electrical
diagram and
(b) sketch.



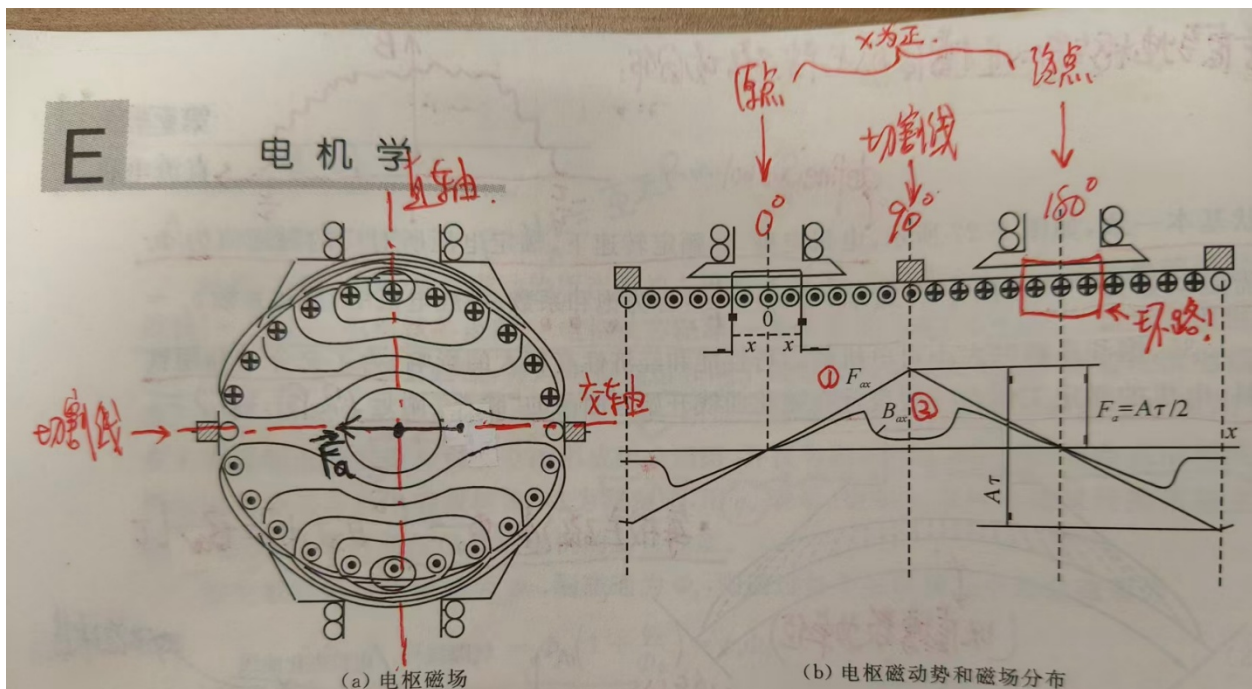


图 2-28 电刷放在几何中性线上时的电枢磁动势和磁场

由此可以画出沿电枢表面电枢磁动势的分布曲线 $F_{ax} = f(x)$, 如图 2-28(b) 所示。从得到气隙电枢磁密 B_{ax} 的空间分布为

$$\textcircled{2} B_{ax} = \mu_0 H_{ax} = \mu_0 \frac{F_{ax}}{\delta} = \frac{\mu_0 A\tau}{\delta} \quad (2-1)$$

显然, 尽管离磁极中心线愈远, 磁动势愈大, 但由于极间气隙增大, 该处的磁密 B 后

应电势方向和电刷电位的正负极性如图 2-18 所示。

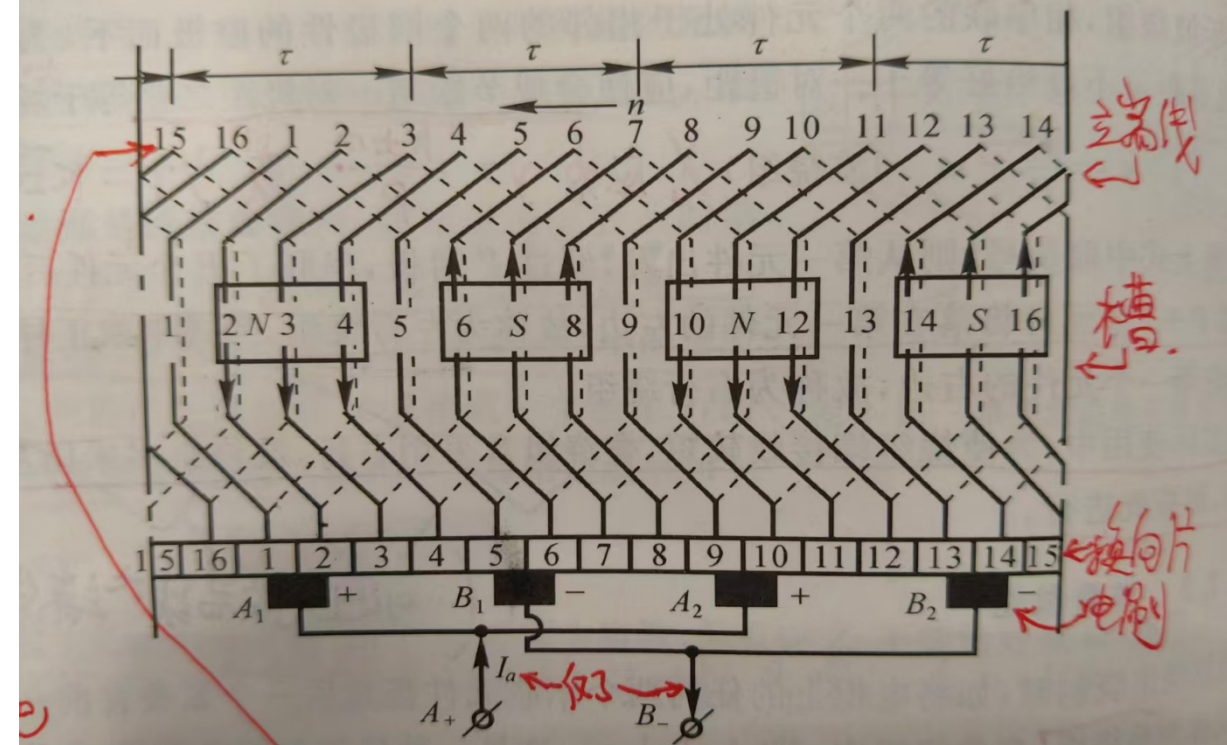


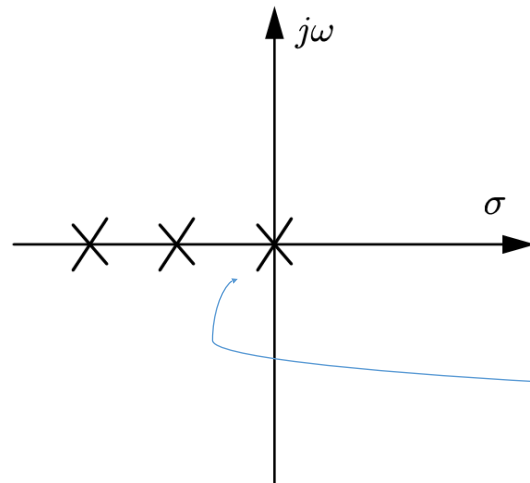
图 2-18 单叠绕组展开图



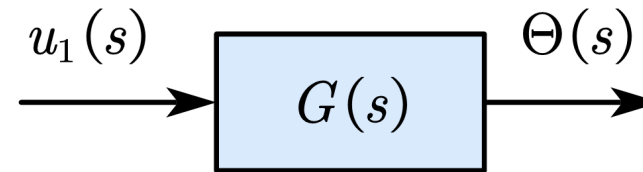
Mathematic Model Overview

Overview of the Main Models

- ❑ O.D.E (from physics laws)
- ❑ Block diagram
- ❑ Transfer function (via Laplace transform)
- ❑ State space model
- ❑ Pole zero map (related to eigenvalues)
 - Root locus and pole placement



Eigenvalues
(when input is set to 0)



Can you give details within $G(s)$?

$$\frac{d}{dt} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & K_T J^{-1} \\ 0 & -K_E L^{-1} & -R L^{-1} \end{bmatrix} \begin{bmatrix} \Theta \\ \Omega \\ i \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ L^{-1} \end{bmatrix} u + \begin{bmatrix} 0 \\ J^{-1} \\ 0 \end{bmatrix} \tau_L$$



Example 2.23

Transfer Function—DC Motor and Load

PROBLEM: Given the system and torque-speed curve of Figure 2.39(a) and (b), find the transfer function, $\theta_L(s)/E_a(s)$.

SOLUTION: Begin by finding the mechanical constants, J_m and D_m , in Eq. (2.153). From Eq. (2.155), the total inertia at the armature of the motor is

$$J_m = J_a + J_L \left(\frac{N_1}{N_2} \right)^2 = 5 + 700 \left(\frac{1}{10} \right)^2 = 12 \quad (2.164)$$

and the total damping at the armature of the motor is

$$D_m = D_a + D_L \left(\frac{N_1}{N_2} \right)^2 = 2 + 800 \left(\frac{1}{10} \right)^2 = 10 \quad (2.165)$$

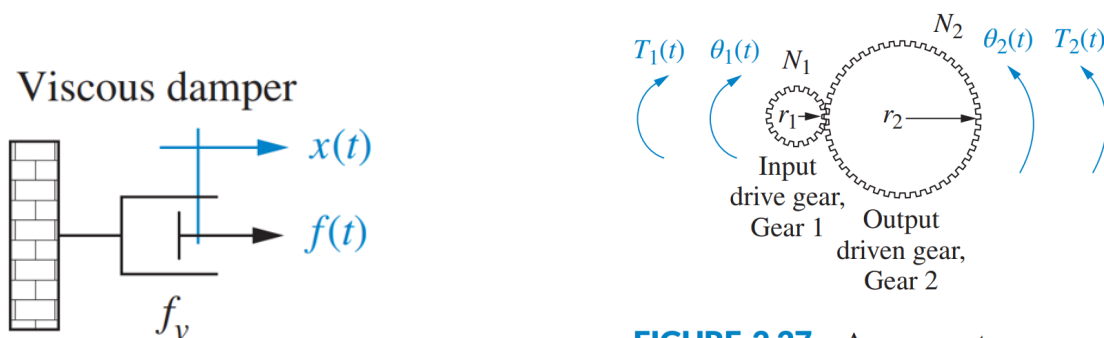


FIGURE 2.27 A gear system

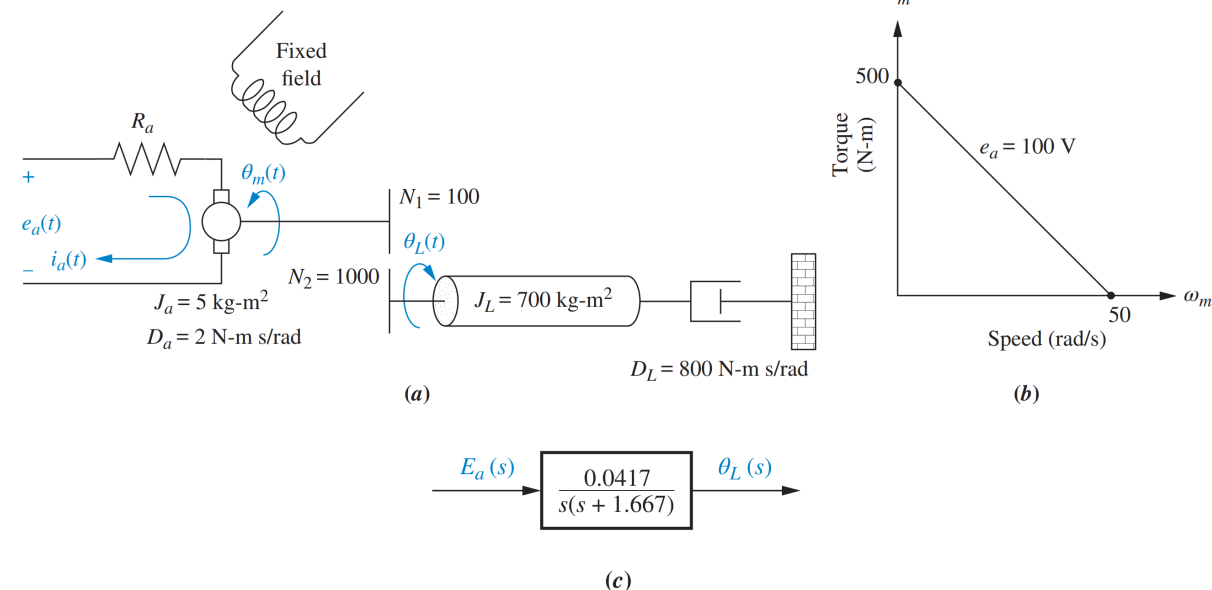


FIGURE 2.39 a. DC motor and load; b. torque-speed curve; c. block diagram

Generalizing the results, we can make the following statement: *Rotational mechanical impedances can be reflected through gear trains by multiplying the mechanical impedance by the ratio*

$$\left(\frac{\text{Number of teeth of gear on destination shaft}}{\text{Number of teeth of gear on source shaft}} \right)^2$$

where the impedance to be reflected is attached to the source shaft and is being reflected to the destination shaft. The next example demonstrates the application of the concept of reflected impedances as we find the transfer function of a rotational mechanical system with gears.



State Space Model

- ❑ The simultaneous **first-order differential** equations about the state variables
 - State variables must be linearly independent; that is, no state variable can be written as a linear combination of the other state variables.

- ❑ **Algebraic** output equation

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$

- ❑ E.g.,:

$\mathbf{x}$ = state vector

$\dot{\mathbf{x}}$ = derivative of the state vector with respect to time

$\mathbf{y}$ = output vector

$\mathbf{u}$ = input or control vector

$\mathbf{A}$ = system matrix

$\mathbf{B}$ = input matrix

$\mathbf{C}$ = output matrix

$\mathbf{D}$ = feedforward matrix

- ❑ State s

les.

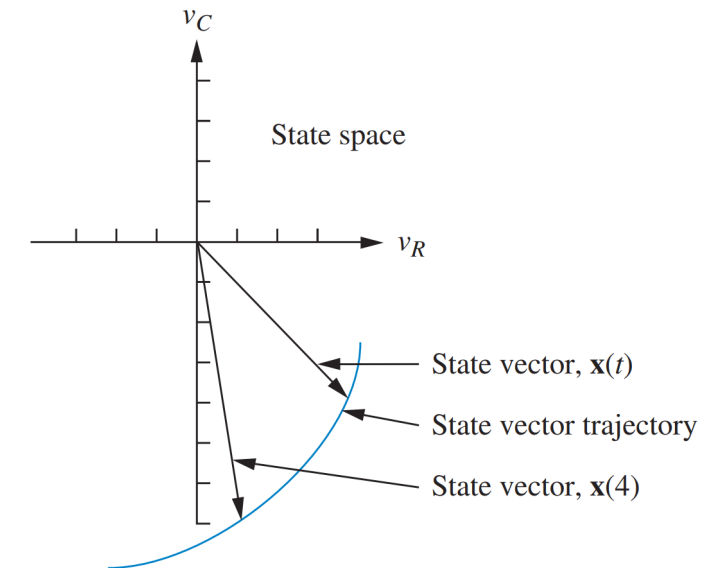


FIGURE 3.3 Graphic representation of state space and a state vector

PROBLEM: Find the state-space representation of the electrical network shown in Figure 3.8. The output is $v_o(t)$.

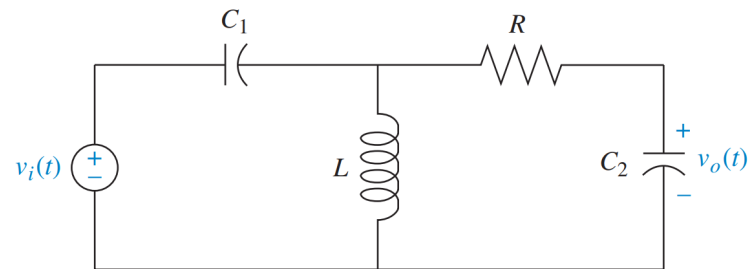


FIGURE 3.8 Electric circuit for Skill-Assessment Exercise 3.1

ANSWER:



PROBLEM: Represent the translational mechanical system shown in Figure 3.9 in state space, where $x_3(t)$ is the output.

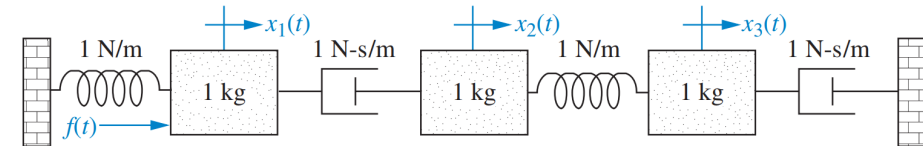
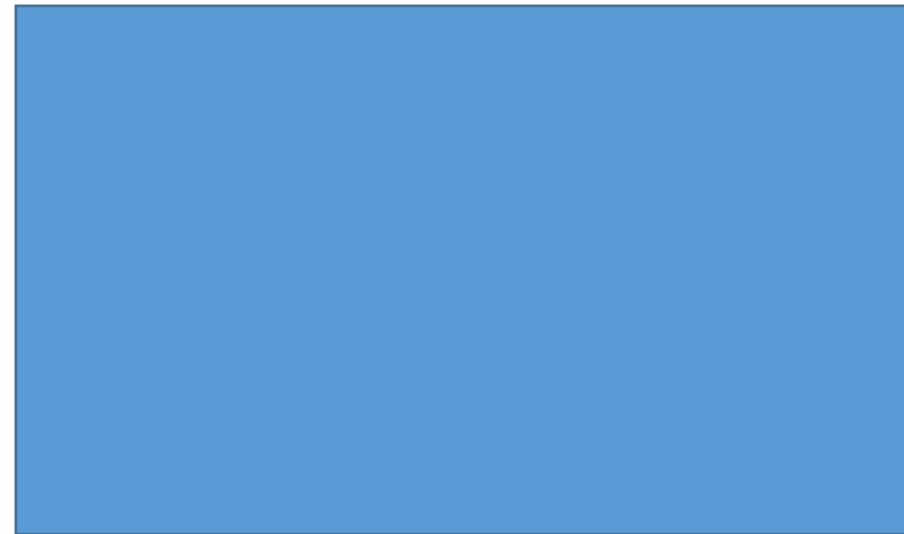


FIGURE 3.9 Translational mechanical system for Skill-Assessment Exercise 3.2

ANSWER:



From O.D.E to State space model



At first we select a set of state variables, called *phase variables*, where each subsequent state variable is defined to be the derivative of the previous state variable.

$$\frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dt^{n-1}} + \cdots + a_1 \frac{dy}{dt} + a_0 y = b_0 u$$



$$\begin{aligned} x_1 &= y \\ x_2 &= \frac{dy}{dt} \\ x_3 &= \frac{d^2 y}{dt^2} \\ &\vdots \\ x_n &= \frac{d^{n-1} y}{dt^{n-1}} \end{aligned}$$



$$\begin{aligned} \dot{x}_1 &= \frac{dy}{dt} \\ \dot{x}_2 &= \frac{d^2 y}{dt^2} \\ \dot{x}_3 &= \frac{d^3 y}{dt^3} \\ &\vdots \\ \dot{x}_n &= \frac{d^n y}{dt^n} \end{aligned}$$



$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= x_3 \\ &\vdots \\ \dot{x}_{n-1} &= x_n \end{aligned}$$



$$\dot{x}_n = -a_0 x_1 - a_1 x_2 \cdots - a_{n-1} x_n + b_0 u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \vdots \\ \dot{x}_{n-1} \\ \dot{x}_n \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & \cdots & 0 \\ \vdots & & & & & & & \\ 0 & 0 & 0 & 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & -a_3 & -a_4 & -a_5 & \cdots & -a_{n-1} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ b_0 \end{bmatrix} u \quad (3.52)$$

$$y = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_{n-1} \\ x_n \end{bmatrix}$$

Converting a Transfer Function with a Constant Term in the Numerator

$$\frac{C(s)}{R(s)} = \frac{24}{s^3 + 9s^2 + 26s + 24}$$

$$(s^3 + 9s^2 + 26s + 24)C(s) = 24R(s)$$

$$\ddot{c} + 9\dot{c} + 26c = 24r$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -24 & -26 & -9 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 24 \end{bmatrix} r$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

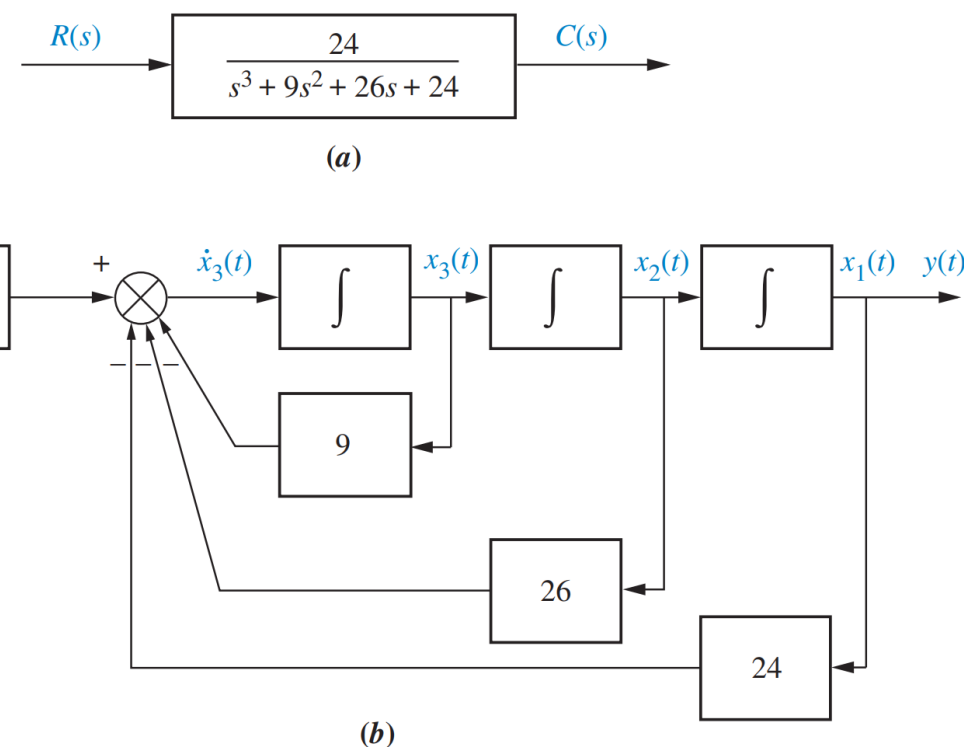


FIGURE 3.10 a. Transfer function; b. equivalent block diagram showing phase variables. Note: $y(t) = c(t)$.

Converting a Transfer Function with a Polynomial in the Numerator

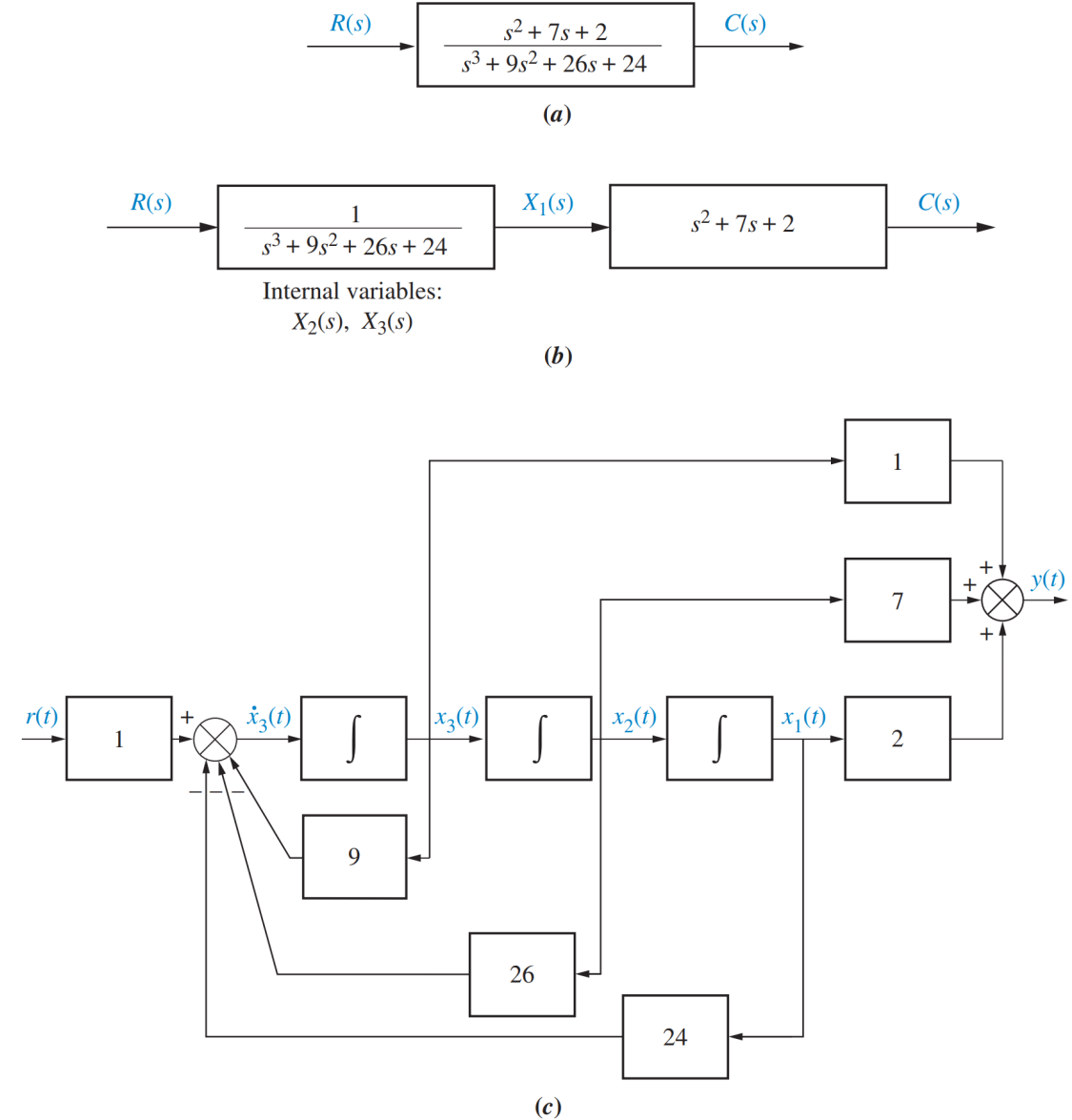
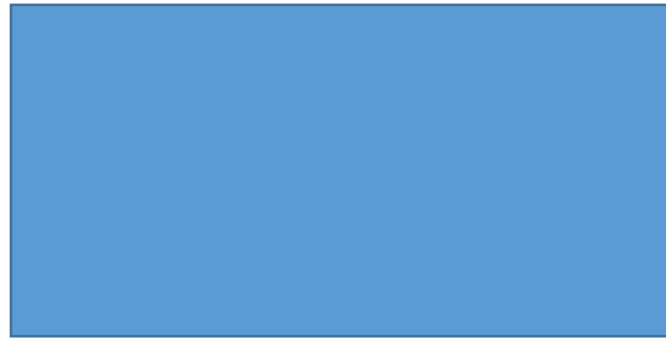


FIGURE 3.12 **a.** Transfer function; **b.** decomposed transfer function; **c.** equivalent block diagram Note: $y(t) = c(t)$.



PROBLEM: Find the state equations and output equation for the phase-variable representation of the transfer function $G(s) = \frac{2s + 1}{s^2 + 7s + 9}$.

ANSWER:



State space model to transfer function (ss2tf)



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x} + \mathbf{D}\mathbf{u}$$



$$s\mathbf{X}(s) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$



$$(s\mathbf{I} - \mathbf{A})\mathbf{X}(s) = \mathbf{B}\mathbf{U}(s)$$



$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s)$$



$$\mathbf{Y}(s) = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) + \mathbf{D}\mathbf{U}(s) = [\mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}]\mathbf{U}(s)$$

Transfer function matrix

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B} + \mathbf{D}$$



Single input single output

State space model to transfer function (ss2tf)



PROBLEM: Given the system defined by Eq. (3.74), find the transfer function, $T(s) = Y(s)/U(s)$, where $U(s)$ is the input and $Y(s)$ is the output.

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 10 \\ 0 \\ 0 \end{bmatrix} u \quad (3.74a)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \mathbf{x} \quad (3.74b)$$

$$T(s) = \frac{Y(s)}{U(s)} = \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \mathbf{B} + \mathbf{D}$$

$$(s\mathbf{I} - \mathbf{A})^{-1} = \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} = \frac{\begin{bmatrix} (s^2 + 3s + 2) & s + 3 & 1 \\ -1 & s(s + 3) & s \\ -s & -(2s + 1) & s^2 \end{bmatrix}}{s^3 + 3s^2 + 2s + 1} \quad (s\mathbf{I} - \mathbf{A}) = \begin{bmatrix} s & 0 & 0 \\ 0 & s & 0 \\ 0 & 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} s & -1 & 0 \\ 0 & s & -1 \\ 1 & 2 & s + 3 \end{bmatrix}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \text{adj}(\mathbf{A}) = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} \\ + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}^T = \begin{bmatrix} + \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} \\ - \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} \\ + \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} & - \begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} & + \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} \end{bmatrix}$$

$$T(s) = \frac{10(s^2 + 3s + 2)}{s^3 + 3s^2 + 2s + 1}$$

TryIt 3.2

Use the following MATLAB and the Control System Toolbox statements to obtain the transfer function shown in Skill-Assessment Exercise 3.4 from the state-space representation of Eq. (3.78).

```
A=[ -4  -1.5; 4  0];  
B=[ 2  0]';  
C=[ 1.5  0.625];  
D=0;  
T=ss(A, B, C, D);  
T=tf(T)
```

PROBLEM: Convert the state and output equations shown in Eq. (3.78) to a transfer function.

$$\dot{\mathbf{x}} = \begin{bmatrix} -4 & -1.5 \\ 4 & 0 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 2 \\ 0 \end{bmatrix} u(t) \quad (3.78a)$$

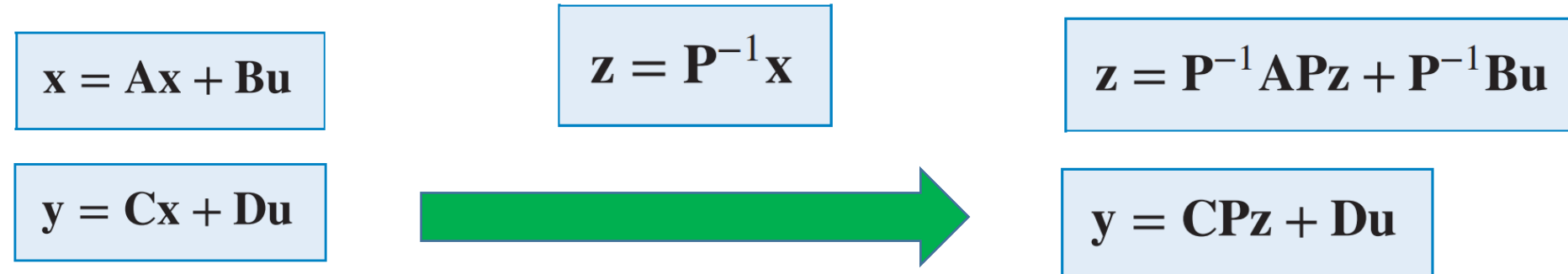
$$y = \begin{bmatrix} 1.5 & 0.625 \end{bmatrix} \mathbf{x} \quad (3.78b)$$

ANSWER:



$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{adj}(\mathbf{A}) = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

State space model is not unique



PROBLEM: Given the system represented in state space by Eqs. (5.73),

$$\dot{\mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u \quad (5.73a)$$

$$y = [1 \quad 0 \quad 0] \mathbf{x} \quad (5.73b)$$

transform the system to a new set of state variables, $\mathbf{z}$, where the new state variables are related to the original state variables, $\mathbf{x}$, as follows:

$$\mathbf{z} = \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \mathbf{x} = \mathbf{P}^{-1} \mathbf{x} \quad \begin{matrix} z_1 = 2x_1 \\ z_2 = 3x_1 + 2x_2 \\ z_3 = x_1 + 4x_2 + 5x_3 \end{matrix} \quad (5.74a)$$

$$z_2 = 3x_1 + 2x_2 \quad (5.74b)$$

$$z_3 = x_1 + 4x_2 + 5x_3 \quad (5.74c)$$

$$\begin{aligned} \mathbf{P}^{-1} \mathbf{A} \mathbf{P} &= \begin{bmatrix} 2 & 0 & 0 \\ 3 & 2 & 0 \\ 1 & 4 & 5 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -2 & -5 & -7 \end{bmatrix} \begin{bmatrix} 0.5 & 0 & 0 \\ -0.75 & 0.5 & 0 \\ 0.5 & -0.4 & 0.2 \end{bmatrix} \\ &= \begin{bmatrix} -1.5 & 1 & 0 \\ -1.25 & 0.7 & 0.4 \\ -2.5 & 0.4 & -6.2 \end{bmatrix} \end{aligned}$$

Diagonal state space representation

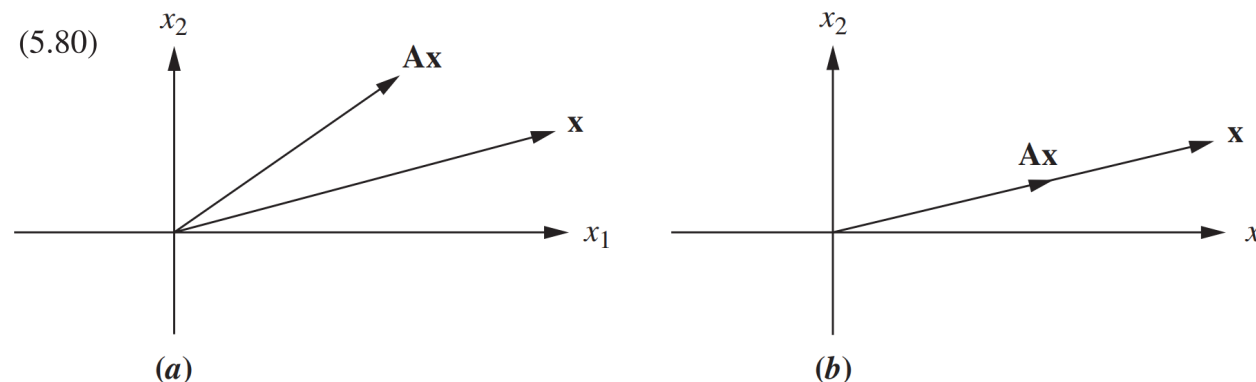


Eigenvector. The eigenvectors of the matrix $\mathbf{A}$ are all vectors, $\mathbf{x}_i \neq \mathbf{0}$, which under the transformation $\mathbf{A}$ become multiples of themselves; that is,

$$\mathbf{A}\mathbf{x}_i = \lambda_i \mathbf{x}_i$$

where λ_i 's are constants.

FIGURE 5.32 To be an eigenvector, the transformation $\mathbf{A}\mathbf{x}$ must be collinear with $\mathbf{x}$; thus, in (a), $\mathbf{x}$ is not an eigenvector; in (b), it is.



Eigenvalue. The eigenvalues of the matrix $\mathbf{A}$ are the values of λ_i that satisfy Eq. (5.80) for $\mathbf{x}_i \neq \mathbf{0}$.

To find the eigenvectors, we rearrange Eq. (5.80). Eigenvectors, $\mathbf{x}_i$, satisfy

$$\mathbf{0} = (\lambda_i \mathbf{I} - \mathbf{A})\mathbf{x}_i \quad (5.81)$$

Solving for $\mathbf{x}_i$ by premultiplying both sides by $(\lambda_i \mathbf{I} - \mathbf{A})^{-1}$ yields

$$\mathbf{x}_i = (\lambda_i \mathbf{I} - \mathbf{A})^{-1} \mathbf{0} = \frac{\text{adj}(\lambda_i \mathbf{I} - \mathbf{A})}{\det(\lambda_i \mathbf{I} - \mathbf{A})} \mathbf{0} \quad (5.82)$$

Since $\mathbf{x}_i \neq \mathbf{0}$, a nonzero solution exists if

$$\det(\lambda_i \mathbf{I} - \mathbf{A}) = 0 \quad (5.83)$$

□ Diagonal representation

- Equivalent to partial fraction expansion of t.f.

$$\mathbf{P} = [\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3, \dots, \mathbf{x}_n]$$

$$\mathbf{A}\mathbf{x}_i = \lambda_i \mathbf{x}_i$$

$$\mathbf{A}\mathbf{P} = \mathbf{P}\mathbf{D}$$

$$\mathbf{D} = \mathbf{P}^{-1} \mathbf{A}\mathbf{P}$$

$$\mathbf{z} = \mathbf{P}^{-1} \mathbf{A}\mathbf{P}\mathbf{z} + \mathbf{P}^{-1} \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{P}\mathbf{z} + \mathbf{D}\mathbf{u}$$

from which λ_i , the eigenvalues, can be found.

Diagonalizing a System in State Space

PROBLEM: Given the system of Eqs. (5.94), find the diagonal system that is similar.

$$\dot{\mathbf{x}} = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 2 \end{bmatrix} u \quad (5.94a)$$

$$y = [2 \quad 3] \mathbf{x} \quad (5.94b)$$

$$\mathbf{P}^{-1} \mathbf{A} \mathbf{P} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix}$$

$$\mathbf{P}^{-1} \mathbf{B} = \begin{bmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix}$$

$$\mathbf{C} \mathbf{P} = [2 \quad 3] \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = [5 \quad -1]$$

$$\dot{\mathbf{z}} = \begin{bmatrix} -2 & 0 \\ 0 & -4 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 3/2 \\ -1/2 \end{bmatrix} u \quad \boxed{\mathbf{z} = \mathbf{P}^{-1} \mathbf{A} \mathbf{P} \mathbf{z} + \mathbf{P}^{-1} \mathbf{B} u}$$

$$y = [5 \quad -1] \mathbf{z} \quad \boxed{y = \mathbf{C} \mathbf{P} \mathbf{z} + \mathbf{D} u}$$

$$\begin{aligned} \det(\lambda \mathbf{I} - \mathbf{A}) &= \left| \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} - \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \right| \\ &= \begin{vmatrix} \lambda + 3 & -1 \\ -1 & \lambda + 3 \end{vmatrix} \\ &= \lambda^2 + 6\lambda + 8 \end{aligned}$$

$$\begin{aligned} \mathbf{A} \mathbf{x}_i &= \lambda \mathbf{x}_i \\ \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= -2 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= -4 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \end{aligned}$$

$$\mathbf{x} = \begin{bmatrix} c \\ c \end{bmatrix}$$

$$\mathbf{x} = \begin{bmatrix} c \\ -c \end{bmatrix}$$

$$\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{x}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Diagonal state space representation



PROBLEM: For the system represented in state space as follows:

$$\dot{\mathbf{x}} = \begin{bmatrix} 1 & 3 \\ -4 & -6 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 1 \\ 3 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 4 \end{bmatrix} \mathbf{x}$$

convert the system to one where the new state vector, $\mathbf{z}$, is

$$\mathbf{z} = \begin{bmatrix} 3 & -2 \\ 1 & -4 \end{bmatrix} \mathbf{x}$$

ANSWER:

$$\dot{\mathbf{z}} = \begin{bmatrix} 6.5 & -8.5 \\ 9.5 & -11.5 \end{bmatrix} \mathbf{z} + \begin{bmatrix} -3 \\ -11 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0.8 & -1.4 \end{bmatrix} \mathbf{z}$$

PROBLEM: For the original system find the diagonal system that is similar.

ANSWER:

$$\dot{\mathbf{z}} = \begin{bmatrix} -2 & 0 \\ 0 & -3 \end{bmatrix} \mathbf{z} + \begin{bmatrix} 18.39 \\ 20 \end{bmatrix} u$$
$$y = \begin{bmatrix} -2.121 & 2.6 \end{bmatrix} \mathbf{z}$$

Solution of the State Equations



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$



$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$



$$\begin{aligned}\mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) \\ &= \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} [\mathbf{x}(0) + \mathbf{B}\mathbf{U}(s)]\end{aligned}$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$

$$\mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}] = \mathcal{L}^{-1}\left[\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}\right] = \Phi(t)$$

$$X(s) = \frac{x(0)}{s - a} + \frac{b}{s - a}U(s).$$

$$x(t) = e^{at}x(0) + \int_0^t e^{+a(t-\tau)}bu(\tau)d\tau.$$

$$\begin{aligned}\mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \\ &= \Phi(t)\mathbf{x}(0) + \int_0^t \Phi(t - \tau)\mathbf{B}\mathbf{u}(\tau)d\tau\end{aligned}$$

zero-input response

Zero-state response as
convolution integral

The *matrix exponential* function is defined by in a similar Taylor series form

$$e^{\mathbf{A}t} = \exp(\mathbf{A}t) = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2t^2}{2!} + \cdots + \frac{\mathbf{A}^kt^k}{k!} + \cdots,$$

Solution of the State Equations



$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{u}$$



$$s\mathbf{X}(s) - \mathbf{x}(0) = \mathbf{A}\mathbf{X}(s) + \mathbf{B}\mathbf{U}(s)$$



$$\begin{aligned}\mathbf{X}(s) &= (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{x}(0) + (s\mathbf{I} - \mathbf{A})^{-1}\mathbf{B}\mathbf{U}(s) \\ &= \frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})} [\mathbf{x}(0) + \mathbf{B}\mathbf{U}(s)]\end{aligned}$$

$$\mathbf{Y}(s) = \mathbf{C}\mathbf{X}(s) + \mathbf{D}\mathbf{U}(s)$$

$$\mathcal{L}^{-1}[(s\mathbf{I} - \mathbf{A})^{-1}] = \mathcal{L}^{-1}\left[\frac{\text{adj}(s\mathbf{I} - \mathbf{A})}{\det(s\mathbf{I} - \mathbf{A})}\right] = \Phi(t)$$

for scalar case

$$X(s) = \frac{x(0)}{s - a} + \frac{b}{s - a}U(s).$$

$$x(t) = e^{at}x(0) + \int_0^t e^{a(t-\tau)}bu(\tau)d\tau.$$

$$\begin{aligned}\mathbf{x}(t) &= e^{\mathbf{A}t}\mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)}\mathbf{B}\mathbf{u}(\tau)d\tau \\ &= \Phi(t)\mathbf{x}(0) + \int_0^t \Phi(t - \tau)\mathbf{B}\mathbf{u}(\tau)d\tau\end{aligned}$$

zero-input response

Zero-state response as
convolution integral

The *matrix exponential* function is defined by in a similar Taylor series form

$$e^{\mathbf{A}t} = \exp(\mathbf{A}t) = \mathbf{I} + \mathbf{A}t + \frac{\mathbf{A}^2t^2}{2!} + \cdots + \frac{\mathbf{A}^kt^k}{k!} + \cdots,$$



Block Diagram

Basic components:

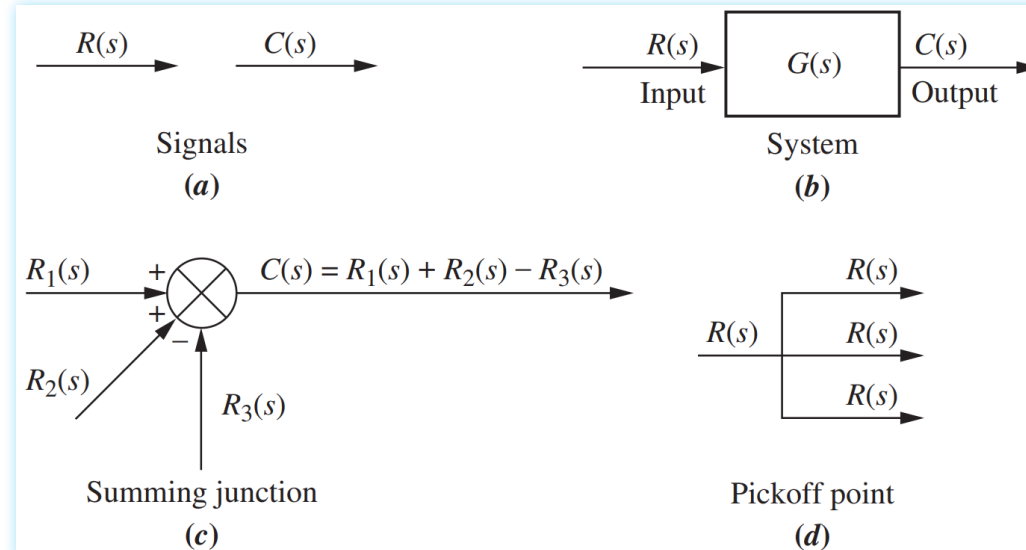


FIGURE 5.2 Components of a block diagram for a linear, time-invariant system

Cascade and parallel:

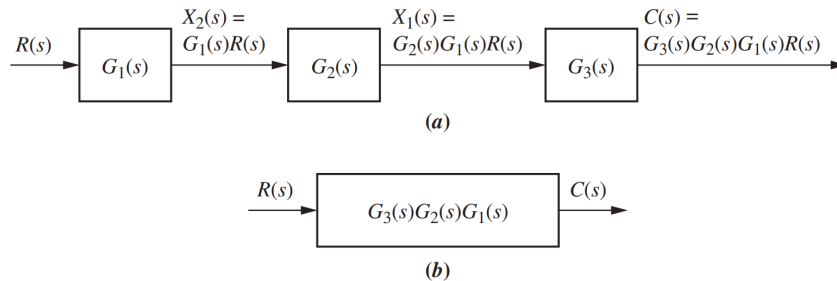


FIGURE 5.3 a. Cascaded subsystems; b. equivalent transfer function

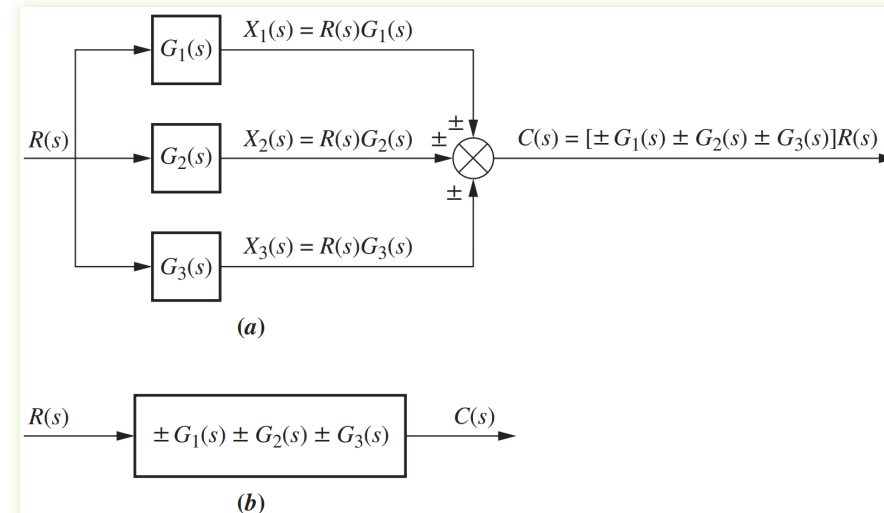


FIGURE 5.5 a. Parallel subsystems; b. equivalent transfer function

Feedback

$$G_e(s) = \frac{G(s)}{1 \pm G(s)H(s)}$$

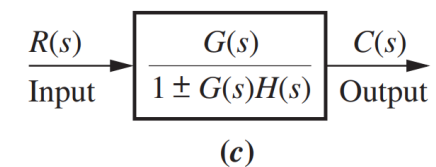
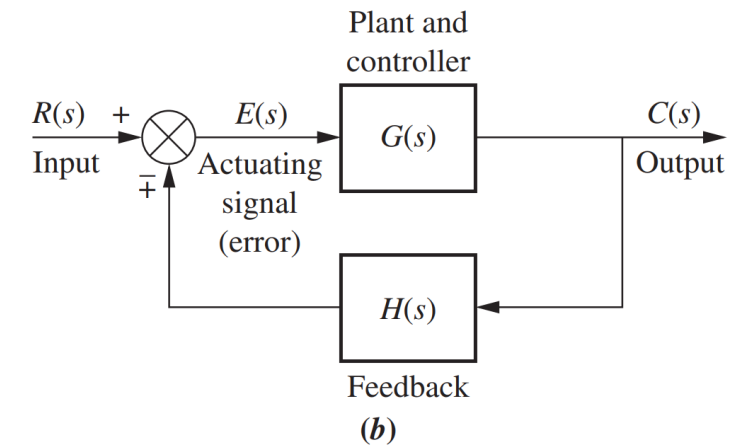
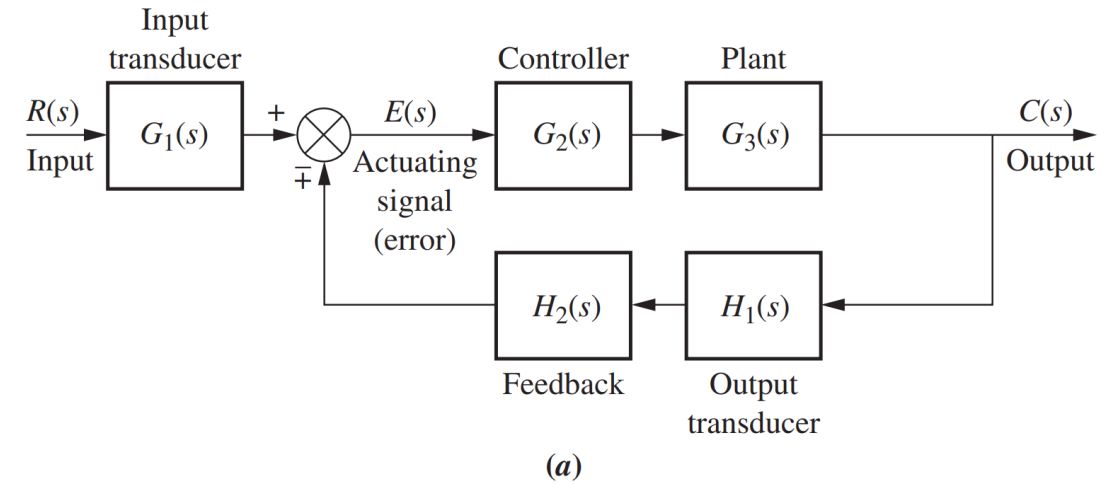


FIGURE 5.6 **a.** Feedback control system; **b.** simplified model; **c.** equivalent transfer function

□ Moving blocks

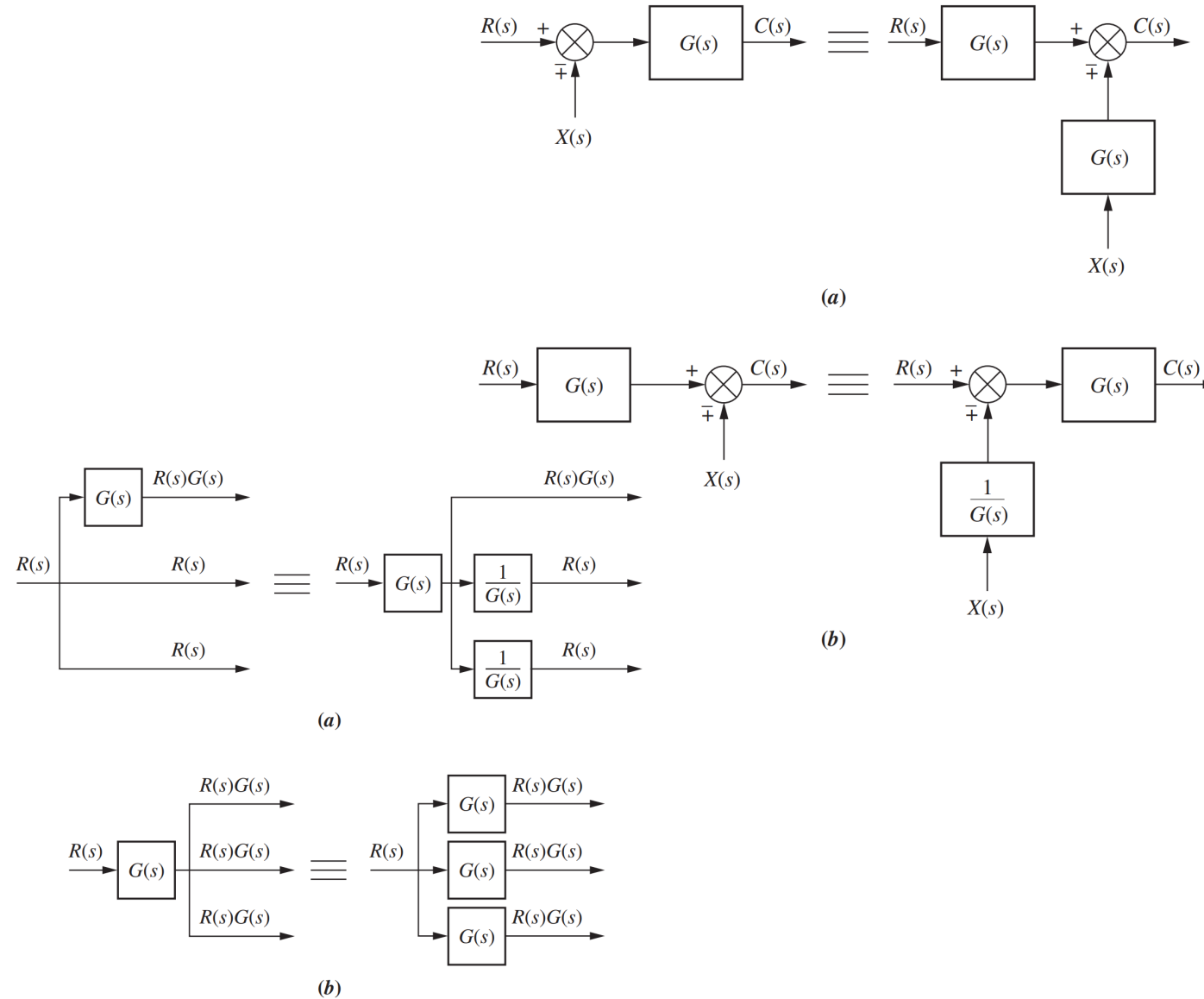


FIGURE 5.7 Block diagram algebra for summing junctions—equivalent forms for moving a block **a.** to the left past a summing junction; **b.** to the right past a summing junction

FIGURE 5.8 Block diagram algebra for pickoff points—equivalent forms for moving a block **a.** to the left past a pickoff point; **b.** to the right past a pickoff point

PROBLEM: Reduce the block diagram shown in Figure 5.9 to a single transfer function.

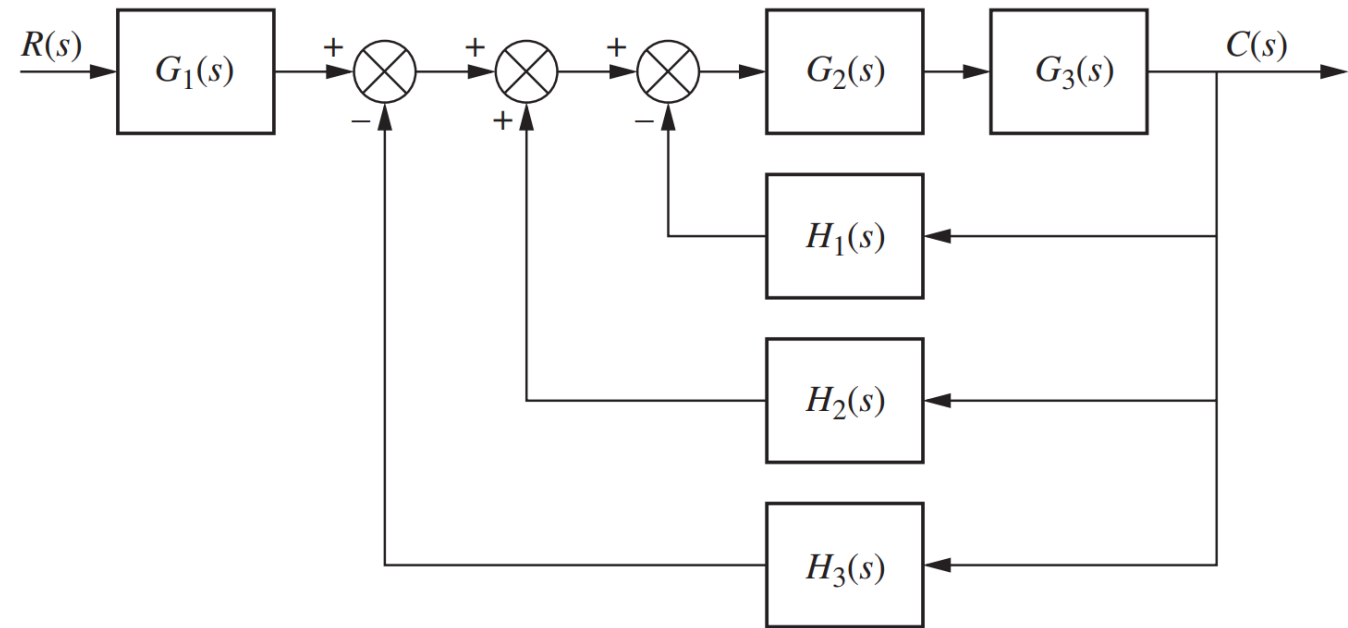
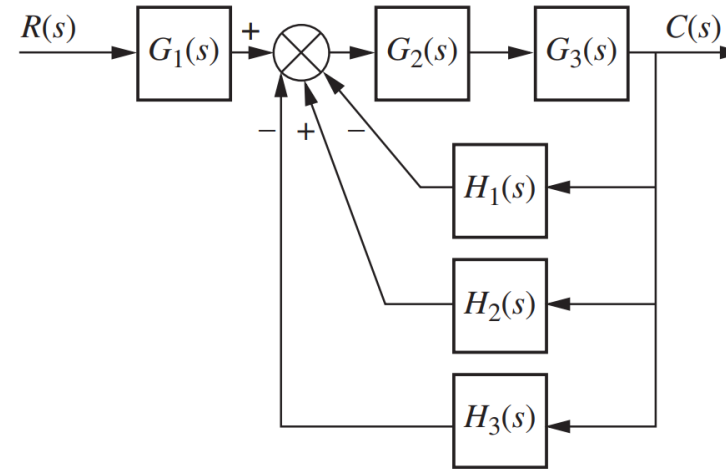


FIGURE 5.9 Block diagram for Example 5.1

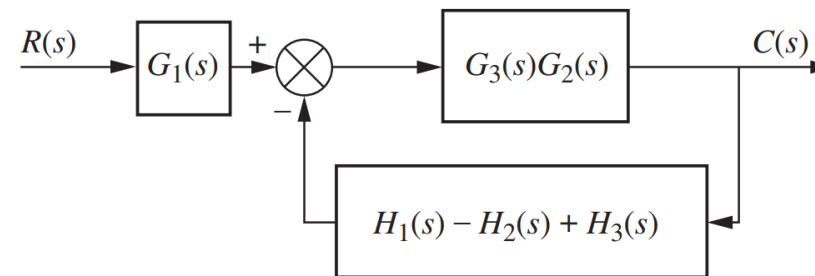
Block diagram reduction



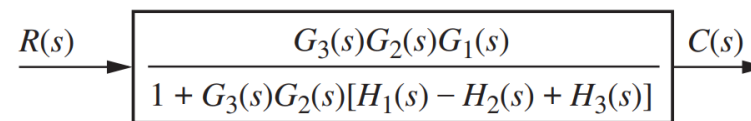
ANSWER:



(a)



(b)



(c)

FIGURE 5.10 Steps in solving Example 5.1:
a. Collapse summing junctions;
b. form equivalent cascaded system in the forward path and equivalent parallel system in the feedback path; **c.** form equivalent feedback system and multiply by cascaded $G_1(s)$

Block Diagram Reduction by Moving Blocks

PROBLEM: Reduce the system shown in Figure 5.11 to a single transfer function.

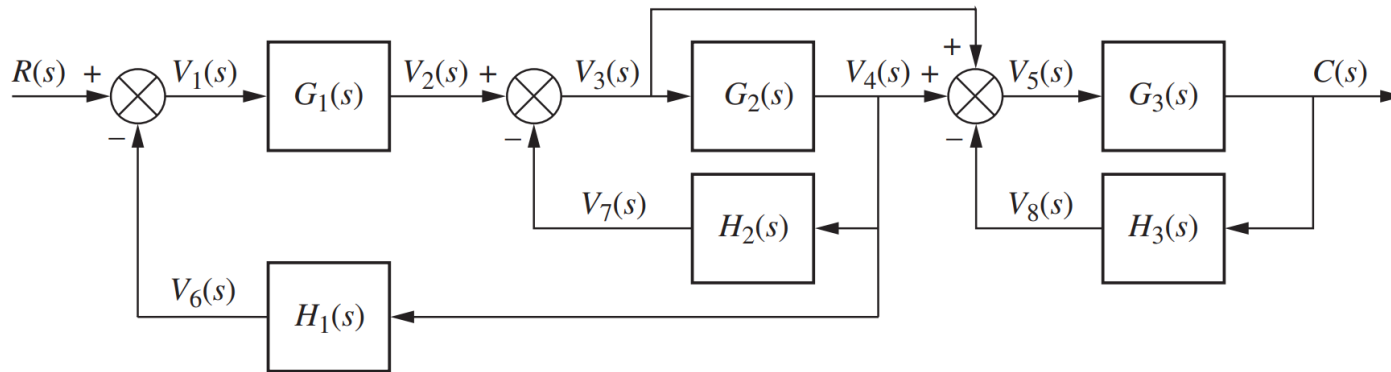
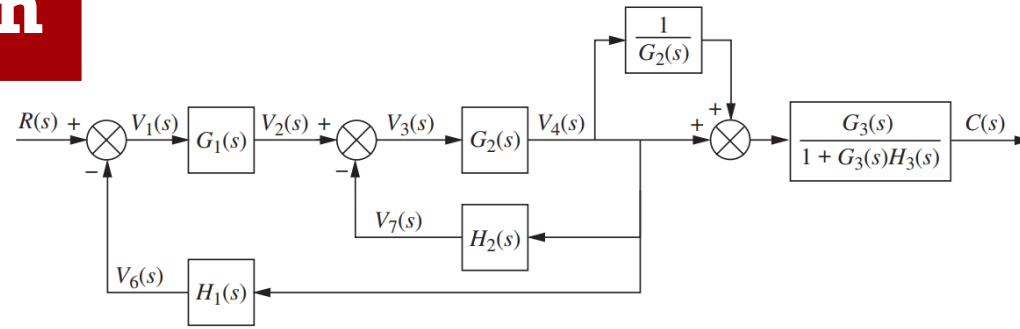


FIGURE 5.11 Block diagram for Example 5.2

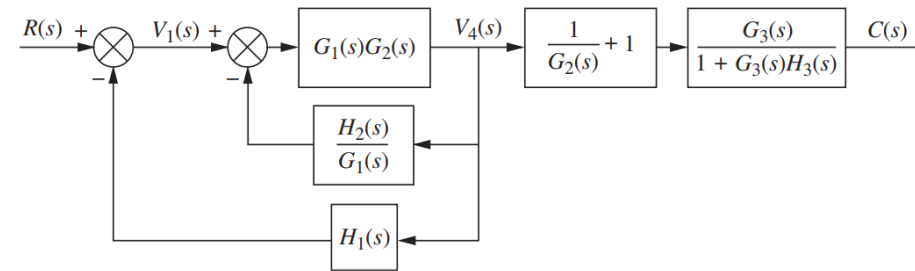
Block diagram reduction



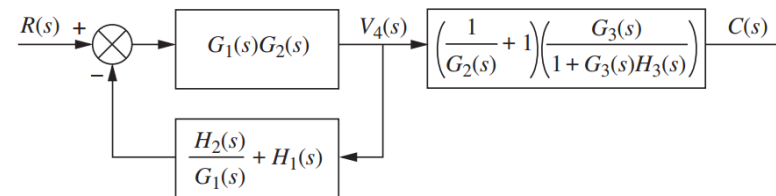
ANSWER:



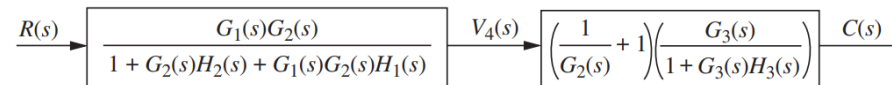
(a)



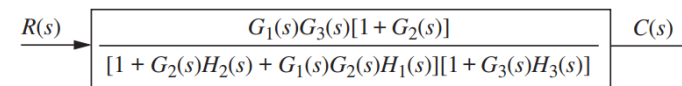
(b)



(c)



(d)



(e)

FIGURE 5.12 Steps in the block diagram reduction for Example 5.2

Block diagram reduction



PROBLEM: Find the equivalent transfer function, $T(s) = C(s)/R(s)$, for the system shown in Figure 5.13.

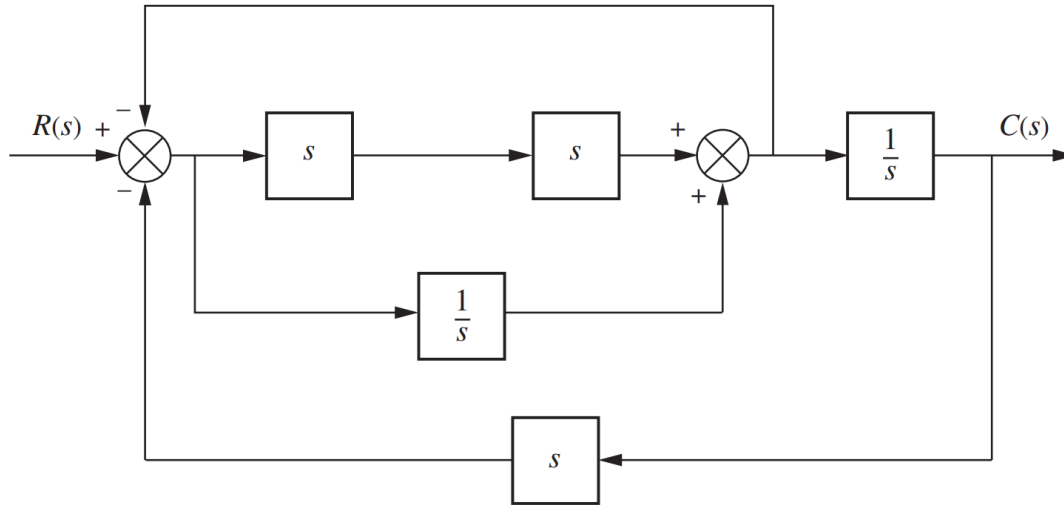


FIGURE 5.13 Block diagram for Skill-Assessment Exercise 5.1

ANSWER:



TryIt 5.1

Use the following MATLAB and Control System Toolbox statements to find the closed-loop transfer function of the system in Example 5.2 if all $G_i(s) = 1/(s + 1)$ and all $H_i(s) = 1/s$.

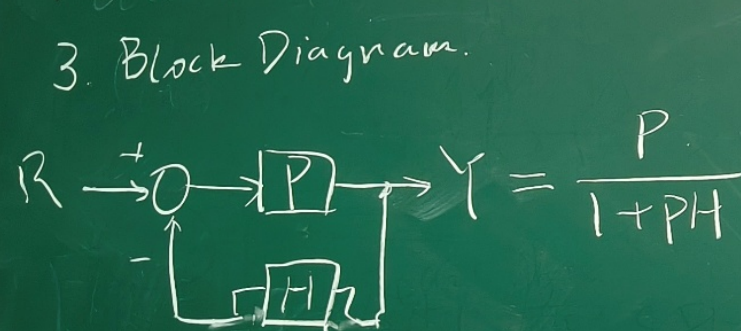
```
G1=tf(1,[1 1]);
G2=G1; G3=G1;
H1=tf(1,[1 0]);
H2=H1; H3=H1;
System=append(...
    (G1, G2, G3, H1, H2, H3);
input=1; output=3;
Q=[ 1  -4   0   0   0
     2   1  -5   0   0
     3   2   1  -5  -6
     4   2   0   0   0
     5   2   0   0   0
     6   3   0   0   0];
T=connect(System,...
    Q, input, output);
T=tf(T); T=minreal(T)
```



Appendix

System Model

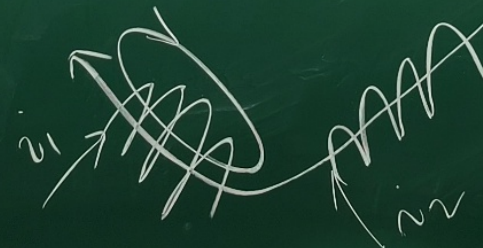
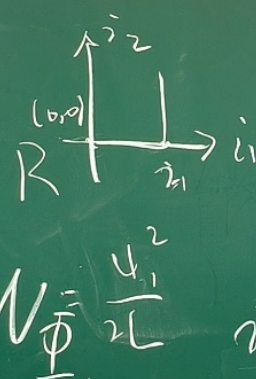
1. O.D.E $\rightarrow$ initial conditions set to zero $\left(\frac{Y(s)}{R(s)} = \frac{S^2}{s+1} \right)$ (not proper)
2. Transfer Function (only describe a pair of I/O relation)
3. Block Diagram.



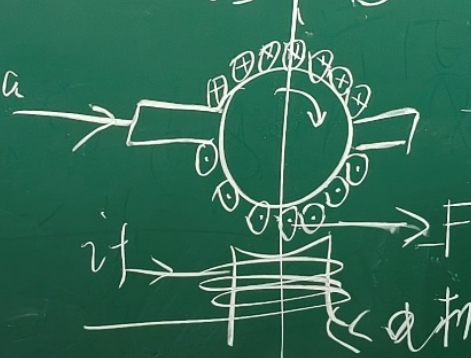
4. Signal Flow Graph

$$d\psi_1(\vec{r}, \vec{m}) = \frac{\partial \psi_1}{\partial \vec{r}} d\vec{r} + \frac{\partial \psi_1}{\partial \vec{m}} d\vec{m}$$

$$\psi_1 = L_1 i_1 + L_2 i_2$$

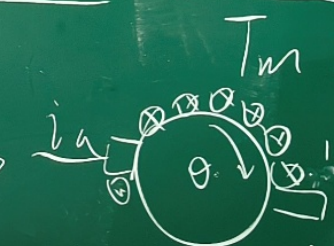


Force-Voltage
Analogy



$$u - Ri = L \frac{di}{dt} \rightarrow \text{Faraday's law}$$

$$T = J \frac{d\omega}{dt} = K_m i_f(t) - b\omega$$



$$e = -\frac{d\psi}{dt} = -\frac{d(Li)}{dt}$$

$$= \frac{dL}{dt} i + L \frac{di}{dt}$$

$$= \frac{\partial L}{\partial \omega} \omega + L \frac{di}{dt}$$

Conclusion: The solution of the state differential equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

is found to be

$$\mathbf{x}(t) = \exp(\mathbf{A}t)\mathbf{x}(0) + \int_0^t \exp[\mathbf{A}(t - \tau)]\mathbf{B}\mathbf{u}(\tau)d\tau.$$

Proof:

- *Uniqueness of Solutions*

If we have two solutions $x_1, x_2 : \mathbb{R} \rightarrow \mathbb{R}^{n_x}$, then $y = x_1 - x_2$ satisfies

$$\dot{y}(t) = Ay(t) \quad \text{with} \quad y(0) = 0.$$

The auxiliary function $v(t) = e^{-At}y(t)$ satisfies

$$\dot{v}(t) = -Ae^{-At}y(t) + e^{-At}Ay(t) = -Ae^{-At}y(t) + Ae^{-At}y(t) = 0$$

$$v(0) = 0,$$

$$\implies v(t) = y(t) = 0 \implies x_1 = x_2.$$

Solution of Differential Equations



Conclusion: The solution of the state differential equation

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t)$$

is found to be

$$\mathbf{x}(t) = \exp(\mathbf{A}t)\mathbf{x}(0) + \int_0^t \exp[\mathbf{A}(t - \tau)]\mathbf{B}\mathbf{u}(\tau)d\tau.$$

Proof:

- Verify the ODE

Generalized Leibniz integral rule.

$$\frac{d}{dt} \int_{a(t)}^{b(t)} g(t, \tau) d\tau = g(t, b(t)) \dot{b}(t) - g(t, a(t)) \dot{a}(t) + \int_{a(t)}^{b(t)} g_t(t, \tau) d\tau .$$

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{A} e^{\mathbf{A}t} \mathbf{x}(0) + e^{\mathbf{A}(t-t)} \mathbf{B} \mathbf{u}(t) + \int_0^t \mathbf{A} e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau \\ &= \mathbf{A} \left[e^{\mathbf{A}t} \mathbf{x}(0) + \int_0^t e^{\mathbf{A}(t-\tau)} \mathbf{B} \mathbf{u}(\tau) d\tau \right] + \mathbf{B} \mathbf{u}(t) \end{aligned}$$

Solution of Differential Equations



Specially, the solution of *an unforced system*

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t)$$

is found to be

$$\mathbf{x}(t) = \exp(\mathbf{A}t)\mathbf{x}(0)$$

The matrix exponential function describes the unforced response of the system and is called *the fundamental or state transition matrix* $\Phi(t, 0)$.

Thus, the general solution can be written as

$$\mathbf{x}(t) = \Phi(t)\mathbf{x}(0) + \int_0^t \Phi(t - \tau)\mathbf{B}\mathbf{u}(\tau)d\tau.$$

NOTE, up to now, we are talking about LTI system, for nonlinear or time-varying, there is NO nice general solution form.